Theoretical Handout: Data-Driven Modeling of Galactic Rotation Curves with Artificial Neural Networks

Predecessor Lineage, Methodological Framework, and Paper Evolution

arXiv:2404.05833v2 Analysis & Reproduction Plan

October 6, 2026

Abstract

This handout provides a comprehensive theoretical overview of the paper *"Data-driven modeling of rotation curves with artificial neural networks"* (arXiv:2404.05833v2) by Gabriela Garcia-Arroyo, Isidro Gómez-Vargas, and J. Alberto Vázquez. It synthesizes the scientific context of galactic rotation curves, traces the methodological lineage from predecessor cosmological machine learning papers, details the evolution from arXiv v1 to v2, presents the mathematical formulation of Monte Carlo Dropout and Genetic Algorithm optimization, and outlines the benchmark classifications of the 17 THINGS galaxies.

1 Scientific Context & Problem Formulation

Galactic rotation curves (RCs) measure the circular velocity $V(r)$ of stars and gas as a function of radial distance r from the galactic center. In traditional astrophysics, rotation curves are modeled parametrically as the quadrature sum of baryonic and dark matter (DM) contributions:

$$V^2(r) = V^2_{\text{gas}}(r) + \Upsilon_\star V^2_\star(r) + V^2_{\text{halo}}(r) \tag{1}$$

where:

- $V_{\text{gas}}(r)$ is derived from 21 cm neutral hydrogen (HI) surface density profiles.
- $V_\star(r)$ is derived from 3.6 μm Spitzer surface brightness profiles $I(r)$.
- $\Upsilon_\star$ is the mass-to-light ratio converting luminosity to gravitational mass.
- $V_{\text{halo}}(r)$ is modeled assuming a dark matter density profile, typically Navarro-Frenk-White (NFW):

$$\rho_{\text{NFW}}(r) = \frac{\rho_i}{\left(\frac{r}{R_s}\right)\left(1+\frac{r}{R_s}\right)^2} \tag{2}$$

1.1 Limitations of Parametric Models

Parametric fits impose strong physical assumptions:

1. Smooth, symmetric matter distributions.
2. Fixed or global mass-to-light ratios ($\Upsilon_\star^{\text{fix}}$ vs. $\Upsilon_\star^{\text{free}}$).
3. Inability to capture localized dynamics, non-circular motions, or irregular baryonic structures.

2 Methodological Lineage & Intellectual Heritage

The artificial neural network (ANN) framework in arXiv:2404.05833v2 is adapted directly from a sequence of cosmological machine learning papers by the same authors:

Methodological Lineage

1. Gómez-Vargas et al. (PRD 2023, 107, 043509):
Introduced the `nnogada` library combining Genetic Algorithms (DEAP framework) and Monte Carlo Dropout for model-independent cosmological data reconstructions ($H(z)$, SNe Ia).

2. Mitra, Gómez-Vargas & Zarikas (Phys. Dark Univ. 2024, 101706):
Applied ANN + MC-DO + GA to joint cosmological datasets, comparing performance against Gaussian Processes (GP).

3. Garcia-Arroyo, Gómez-Vargas & Vázquez (arXiv:2404.05833v2 2026):
Transferred the validated cosmological pipeline to **galactic dynamics**, replacing redshift z with radius r, and predicting velocity $V(r)$ and uncertainty $\sigma_V(r)$.

3 Evolution: arXiv v1 vs arXiv v2

Feature	**arXiv v1 (April 2024)**	**arXiv v2 (Feb 2026)**
Title	*Reconstructing rotation curves with artificial neural networks*	*Data-driven modeling of rotation curves with artificial neural networks*
Network Output	Described primarily as single velocity prediction	Multi-output coupled prediction: input $r \to [V(r), \sigma_V(r)]$
LOESS Comparison	Brief citation to LOESS+SIMEX	Expanded discussion highlighting global ANN mapping vs. local piecewise polynomials
Galaxy Grouping	Preliminary MSE reporting	Detailed physical categorization into Groups A, B, and C

Table 1: Comparison of key revisions between v1 and v2.

4 Machine Learning Framework

4.1 Multi-Output Neural Network Topology

For each galaxy, a dedicated feed-forward network maps radial distance $r \to [V(r), \sigma_V(r)]$ with ReLU activations in hidden layers and linear outputs.

4.2 Monte Carlo Dropout (MC-DO)

Dropout with rate $p = 0.3$ remains active during inference. Performing N_{MC} stochastic passes yields:

$$\bar{V}(r) = \frac{1}{N_{\mathrm{MC}}} \sum_{t=1}^{N_{\mathrm{MC}}} \hat{V}^{(t)}(r) \tag{3}$$

$$\sigma^2_{\mathrm{MC}}(r) = \frac{1}{N_{\mathrm{MC}}} \sum_{t=1}^{N_{\mathrm{MC}}} \left(\hat{V}^{(t)}(r) - \bar{V}(r) \right)^2 \tag{4}$$

$$\sigma_{\mathrm{total}}(r) = \sqrt{\sigma^2_{\mathrm{MC}}(r) + \sigma^2_{\mathrm{stat}}(r)} \tag{5}$$

4.3 Genetic Algorithm Search Space

Using `nnogada`, the GA optimizes:

- Number of hidden layers $\in [3, 4]$
- Neurons per hidden layer $\in [50, 100, 150, 200]$
- Batch size $\in [4, 8, 16, 32]$

Fixed training parameters: Adam optimizer ($\mathrm{lr} = 5 \times 10^{-4}$), 1000 epochs, MSE loss.

5 Benchmark Galaxy Architectures (Table 1 Extraction)

Galaxy	# Hidden Layers	# Neurons	Batch Size
NGC4736	4	200	8
NGC2841	3	150	8
NGC925	4	200	4
NGC2903	4	150	8
DDO154 (Target)	**4**	**200**	**4**
NGC6946	4	200	8
NGC7793	4	150	4
NGC2976	4	200	4
NGC7331	3	200	4
NGC3198	3	150	4
NGC3621	4	200	4
NGC3521	3	200	4
NGC5055	4	100	4
NGC2366	4	200	8
NGC2403	3	150	8
NGC3031	3	200	4
IC2574	3	200	4

Table 2: Best hyperparameter combinations extracted from Table 1 of arXiv:2404.05833v2.